SCHEME OF VALUATION / ANSWER KEY

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S3 (R) (FT/WP) Examination November 2025 (2024 Scheme)

Course Code: GAEST305

Course Name: DIGITAL ELECTRONICS AND LOGIC DESIGN

Note: The Question paper was lengthy, and hence liberal valuation is expected

Max. Marks: 60			Duration: 2hours30minutes

PART A

(Answer all questions. Each question carries 3 marks) — CO — Marks

1 (CO 1) (3)

i. $39.625_{10} = 100111.101_2$

ii. $A3.B2_{16} = 163.6953125_{10}$

(1.5 marks each)

2 (CO 3) (3)

Either a flowchart or a proper explanation (3 marks)

3 (CO 2) (3)

Truth table – 1.5 mark. Expression – 1.5 mark.

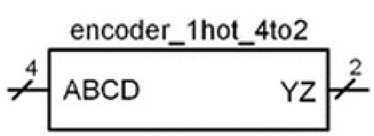
Logic circuit – Not mandatory

$F = A'BC + AB'C + ABC'$

4 (CO 3) (3)

```
assign #5 H = T && E;
```

5 (CO 4) (3)



ABCD	YZ
"0001"	"00"
"0010"	"01"
"0100"	"10"
"1000"	"11"

Y = A + B Z = A + C

Truth table – 1.5 mark. Expression – 2 mark.

6 (CO 4) (3)

```
module xor_gate (output Y, input A, B);
  wire n1, n2, n3;
  nand (n1, A, B);
  nand (n2, A, n1);
  nand (n3, B, n1);
  nand (Y, n2, n3);
endmodule
```

OR

```
module xor_gate (output Y, input A, B);
  wire n1, n2,notA,notB;
not(notA,A);
not(notB,B);
  nand (n1, notA, B);
  nand (n2, A, notB);
  nand (Y, n1, n2);
```

		`endmodule` **Give full marks for any correct implementation.(only Verilog code required)**		
7		Building blocks – State register/state memory, next state logic, output logic.(1 marks) Either 2 differences or Block diagrams- 2 marks	5	(3)

8		`module and_gate (output reg Y, input A, B);`
		`  always @(*) begin`
		`    Y = A & B;`
		`  end`
		`endmodule`
			5	(3)

PART B				
(Answer any one full question from each module, each question carries 9 marks)				
Module -1				
9	a)	Comparison the fixed-point and floating-point number (1.5marks) range and precision of numbers (2 marks) example for each representation (1.5 marks)	1	(5)
	b)	Range of voltages considered as logic high and low. (1 marks) Noise margin- measure of how much unwanted noise or voltage disturbance a logic signal can tolerate without misinterpreting its logic state. (1.5 marks) Ensure error-free digital operation even in the presence of small noise. (1.5 marks)	2	(4)
10	a)	A – B = Binary representation of A + 2's complement representation of B (2 marks) Decimal representation of result – 1 mark A + B = Binary representation of A + Binary representation of B (1 mark) Decimal representation of result – 1 mark	1	(5)
	b)	Fan out - The maximum number of standard logic gate inputs that a gate output can drive while maintaining valid logic levels (0 and 1). (1 mark) Driving resistive loads - Increases current demand on the output; may reduce voltage levels and increase power dissipation. (1.5 marks) Driving multiple gate inputs - too many inputs cause loading; slower rise/fall times (1.5 marks)	2	(4)
Module -2				
11	a)	Truth Table (2.5 mark) $X_0 = \overline{B_0}$ $X_1 = \overline{B_1}\,\overline{B_0} + B_1 B_0$ $X_2 = \overline{B_2}\,B_1 + \overline{B_2}\,B_0 + B_2\,\overline{B_1}\,\overline{B_0}$ $X_3 = B_3 + B_2 B_1 + B_2 B_0$ (K-map reduction – 2.5 marks) **Circuit diagram – Not mandatory**	2	(5)
	b)	The simplified expression is: F =(A+B) + (B. C +D+AC). **Marks to be given based on the question interpreted by the student**(it was not clear in the question paper)	2	(4)
12	a)	**Full marks to be given if the K-Map is solved correctly** Canonical POS: $F = \Pi M(0,3,4,6)$ $= (A+B+C)\,(A+B'+C')\,(A'+B+C)\,(A'+B'+C)$ (1.5marks)	2	(5)

Simplified POS:

$F = (B + C)(A' + C)(A + B' + C')$ (2 marks)

Canonical SOP:

$F = A'B'C + A'BC' + AB'C + ABC'$ (1.5marks)

b) | 3 | (4)

$g_3 = b_3$

$g_2 = b_3 \oplus b_2$

$g_1 = b_2 \oplus b_1$

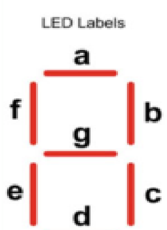
$g_0 = b_1 \oplus b_0$ (1 mark)

```
module binary_to_gray (
    input  [3:0] b,
    output [3:0] g
);
    assign g[3] = b[3];
    assign g[2] = b[3] ^ b[2];
    assign g[1] = b[2] ^ b[1];
    assign g[0] = b[1] ^ b[0];
endmodule
```

(3 marks)

Module -3

13 a) | 4 | (5)



A	B	C	F_a	F_b	F_c	F_d	F_e	F_f	F_g
0	0	0	1	1	1	1	1	1	0
0	0	1	0	1	1	0	0	0	0
0	1	0	1	1	0	1	1	0	1
0	1	1	1	1	1	1	0	0	1
1	0	0	0	1	1	0	0	1	1
1	0	1	1	0	1	1	0	1	1
1	1	0	1	0	1	1	1	1	1
1	1	1	1	1	1	0	0	0	0

(2 marks)

$Fa = A'\cdot C' + B + A\cdot C$

$Fb = B'\cdot C' + A' + B\cdot C$

$Fc = A + B' + C$

$Fd = A'\cdot C' + A'\cdot B + B\cdot C' + A\cdot B'\cdot C$

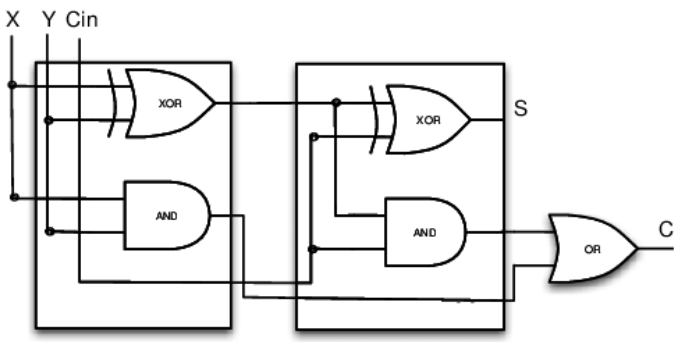
$Fe = A'\cdot C' + B\cdot C'$

$Ff = B'\cdot C' + A\cdot C' + A\cdot B'$

$Fg = A'\cdot B + A\cdot C' + A\cdot B'$

(3 marks)

Full marks to be given if truth table and the K-Map is solved correctly. (Liberal valuation needed as the Answer is lengthy)

<table>
<tr><td></td><td>b)</td><td>Input connections:
$I_0 = \bar{C}$ $I_1 = C$ $I_2 = 0$ $I_3 = 1$ (3 marks)
Multiplexer diagram (1 marks)</td><td>4</td><td>(4)</td></tr>
<tr><td>14</td><td>a)</td><td>
<table>
<tr><th>Component</th><th>Inputs</th><th>Outputs</th><th>Boolean Expression</th></tr>
<tr><td>HA1</td><td>A, B</td><td>S_1, C_1</td><td>$S_1 = A \oplus B, C_1 = A \cdot B$</td></tr>
<tr><td>HA2</td><td>S_1, C_{in}</td><td>$S_2\ (= S), C_2$</td><td>$S_2 = S_1 \oplus C_{in}, C_2 = S_1 \cdot C_{in}$</td></tr>
<tr><td>OR Gate</td><td>C_1, C_2</td><td>C_{out}</td><td>$C_{out} = C_1 + C_2$</td></tr>
</table>


(2 marks)
$S = S_1 \oplus C_{in} = (A \oplus B) \oplus C_{in}$
$C_{out} = (A \cdot B) + ((A \oplus B) \cdot C_{in})$ (2 marks)
Full marks to be given if truth table and the Boolean expressions are correct</td><td>4</td><td>(4)</td></tr>
<tr><td></td><td>b)</td><td>A digital magnitude comparator circuit to compare the 2-bit outputs of sensor S_1 and sensor S_2 is required.
$\mathbf{F} = \mathbf{S_{11}\bar{S}_{21} + S_{10}\bar{S}_{21}\bar{S}_{20} + S_{11}S_{10}\bar{S}_{20}}$
$\mathbf{H} = \mathbf{\bar{S}_{11}S_{21} + \bar{S}_{10}\bar{S}_{11}S_{20} + \bar{S}_{10}S_{21}S_{20}}$
(3 marks)
Truth Table (2 marks)
Circuit diagram is not mandatory.</td><td>4</td><td>(5)</td></tr>
<tr><td colspan="5">Module -4</td></tr>
<tr><td>15</td><td>a)</td><td>Circuit diagram (1 mark each), Timing Diagram – 1 mark each.</td><td>5</td><td>(4)</td></tr>
<tr><td></td><td>b)</td><td>Design (5 marks), Circuit diagram is not mandatory</td><td>5</td><td>(5)</td></tr>
<tr><td>16</td><td>a)</td><td>States (Q2 Q1 Q0) of mod-6 synchronous counter:
$\rightarrow 000 \rightarrow 001 \rightarrow 010 \rightarrow 011 \rightarrow 100 \rightarrow 101$
Design (5 marks), Circuit diagram not mandatory
Give full marks if the counters are designed following the FSM design process using D flip flops.</td><td>5</td><td>(5)</td></tr>
</table>

	b)	See code below. Code (2.5 marks), timing diagram (1.5 marks)	5	(4)

```
module D_FF_Enabled (
    input wire CLK,
    input wire EN,
    input wire D,
    output reg Q
);
always @(posedge CLK) begin
    // The output Q only updates when the clock rises (posedge CLK)
    // AND the enable signal (EN) is HIGH (1).
    if (EN) begin
        Q <= D; // Non-blocking assignment for sequential logic
    end
    // If EN is 0, Q holds its current value (Q <= Q is implied)
end
endmodule
```

Code (2.5 marks), timing diagram (1.5 marks)

Note: The Question paper was lengthy, and hence liberal valuation is expected